Baseline Models Report:
Shifting Tornado Hazard in the United States and Emerging Insurance Exposure

Merrimack College
Data Science Capstone
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1 Background and Questions

This project examines whether tornado hazard in the United States is shifting from the traditional Great Plains Tornado Alley toward the Southeast's Dixie Alley region over time, and whether counties predicted to exhibit elevated tornado hazard in the following year also have greater present-day insurance exposure based on population, housing units and median home value. In this context, insurance exposure refers to the potential for financial loss an insurer may face depending on how many active policies are in the area approximated using population, housing units and median home value.

The hypothesis is that tornado hazard is shifting from the Tornado Alley region toward Dixie Alley due to increasing tornado frequency and severity in the Southeast over time. It is predicted that counties located in Dixie Alley will be more likely classified as high tornado hazard in the following year compared to Tornado Alley and will exhibit a higher present-day insurance exposure.

2 Introduction

There are two major regions in the United States that are commonly associated with elevated tornado activity: the Southeast's region of "Dixie Alley", which includes Arkansas, Louisiana, Mississippi, Alabama, Tennessee and Georgia, and the Great Plains region of "Tornado Alley", which includes Texas, Oklahoma, Kansas and Nebraska. Recent research indicates that tornado activity may be shifting geographically while increasing in frequency in Dixie Alley and decreasing in frequency in Tornado Alley (Gensisi and Brooks, 2018).

This capstone project asks whether tornado hazard in the United States is shifting towards Dixie Alley over time, and whether counties indicating elevated tornado hazard also pose a higher risk of insurance exposure based on population, housing units and median home value.

This is motivated by the importance of understanding shifting trends in tornado risk through an insurance lens. If tornado activity is becoming more frequent and severe in areas with higher housing and population density, like Dixie Alley, insurance companies may be faced with an increased exposure and loss potential. Though the idea of this shift in tornado history has been previously explored, this project provides a specific insurance focus for a predictive modeling framework. It will ultimately examine whether elevated tornado hazard is shifting towards Dixie Alley and occurring in counties with greater insurance exposure.

County-year is the selected unit of analysis for the tornado hazard in this project due to the spatial and temporal component. It's valuable for insurance stakeholders to not only understand the counties with high tornado hazard, but also how that hazard is changing over time. Using county-year observations allows for a deeper analysis into which counties more frequently flag as high hazard, which are becoming more hazardous over time, and ultimately, whether these patterns support a shift towards Dixie Alley potentially representing a higher exposure for insurance companies.

3 Methods

The initial preprocessing plan involved using the cleaned dataset of county-year hazard data for predictive modeling. However, after performing the Exploratory Data Analysis (EDA), multiple adjustments were necessary. Several continuous variables were right-skewed, including population, housing units and property loss. Additionally, the correlation matrix showed overlapping between variables, such as population and housing, indicating multicollinearity. Population and housing show high multicollinearity but as both are valuable to this project due to their representation of different types of insurance exposure (people/policies and property) they will both remain, however, they may impact the coefficient stability.

Additionally, the originally planned construction of the target variable included variables that were also meant to be predictors, which caused concern for data leakage and needed to be reevaluated. To address this, the approach was adjusted so the composite hazard score was created using tornado frequency, intensity and property loss variables while the predictor variables used population, housing units, median home value and tornado characteristics (width and length).

Logistic Regression was selected as the base model due to its interpretability and ability to support the binary classification of the target variable. Additionally, random forest and XGBoost were utilized as flexible models to analyze if performance improves when compared to logistic regression.

Model performance was evaluated using recall and Precision-Recall Area Under the Curve, PR-AUC. Recall was prioritized as the primary evaluation metric to ensure identification of high-risk counties. This is essential for the stakeholder as failure to accurately identify high-risk counties could contribute to the continuation of underwriting in high insurance exposure areas. Confusion matrices also provided a visualization of model performance.

The dataset was split using a time-based approach for county-year observations. The training set included the earlier years of 1979-2014 and the testing set included the later years of 2015-2024. This helped to ensure the model was evaluated on future observations, aligning with the forecasting goal of this project.

Overfitting controls included the time-based train/test split, cross validation and hyperparameter tuning. The time-based split included training years (1979-2014) and test years (2015-2024) to prevent data leakage. A stratified k-fold (5) cross-validation was applied to the training set only to address overfitting, ensuring that observations from both Tornado Alley and Dixie Alley were represented in each fold. Hyperparameter tuning was conducted using randomized search to optimize the model performance and reduce the risk of overfitting.

The model assumptions for logistic regression include linearity between predictors and the log odds of the dependent variable, independence and the absence of perfect multicollinearity while data is collected randomly (Model 5, Models & Assumptions) (Geeks for Geeks, 2025). Multicollinearity was addressed with a correlation matrix which did indicate correlation between some variables, however, they were retained due to their necessity to the dataset for insurance exposure. This may introduce coefficient instability. Additionally, skewed variables for population and housing units were addressed by completing a log transformation. Property loss, which was also highly right-skewed, contributed to the composite hazard score for the target variable and was log transformed and standardized with z-scores to address log-odds. Lastly, a time-based split was conducted to avoid temporal leakage and support independence over time.

Random forest and XGBoost are more flexible models. Multicollinearity impacts these as well. Similarly to logistic regression, they also assume that the data was collected randomly and independently (Model 5, Models & Assumptions) (Geeks for Geeks, 2025).

To address assumptions for each model, multicollinearity was reviewed utilizing the correlation matrix. Though this did show multicollinearity was observed with some variables, such as population and housing, these were not deemed necessary to remove. Linearity was reviewed utilizing log transformations for highly right-skewed variables. For random forest and XGBoost, as they do not rely on linearity but are susceptible to overfitting, this was addressed with cross-validation and hyperparameter tuning.

As the data contains strong geographic structure, there is a risk that the model may learn spatial dependence across counties rather than relying on the intended variables and exposure variables. This will not be fully addressed in this capstone, however, future work could include incorporate spatial lagging of predictors to better account for geographic relationships.

4 Feature Importance Results

Feature importance was addressed for each model. The logistic regression model looked to coefficient magnitude with L1 Regularization (Lasso). Interpretations include that the region categories for Dixie Alley showed a strong positive relationship in addition to the tornado characteristics for average length and width. Housing unit estimate and region category for Tornado Alley had showed a positive relationship as well while median value estimate showed a negative relationship and population estimate dropped to zero.

```
8 x 1 sparse Matrix of class "dgCMatrix"
                                s1
(Intercept)          -7.335385e-01
population_estimate   .
housing_unit_estimate  2.087875e-07
medvalue_estimate    -4.679075e-07
region_categoryT       6.182240e-02
region_categoryD       4.212127e-01
avg_length             4.057942e-03
avg_width              4.699054e-04
```

Image 1: Feature Importance of the Logistic Regression model utilizing L1 Regularization (Lasso)

Random forest utilized mean decrease gini importance. From these results in order of importance from highest to least include: average length, median value, population, housing unit, average width and region category.

	NoRisk	HighRisk	MeanDecreaseAccuracy	MeanDecreaseGini
population_estimate	31.48353	-16.164040	39.31691	1856.3958
housing_unit_estimate	30.03063	-15.144070	37.23014	1854.6738
medvalue_estimate	14.13528	19.812508	28.74519	1903.6892
region_category	29.83333	16.877111	37.03238	217.2282
avg_length	22.60301	-5.188292	18.46292	2088.9699
avg_width	18.60400	4.348705	20.46235	1775.0096

Image 2: Feature Importance of the random forest model utilizing gini importance

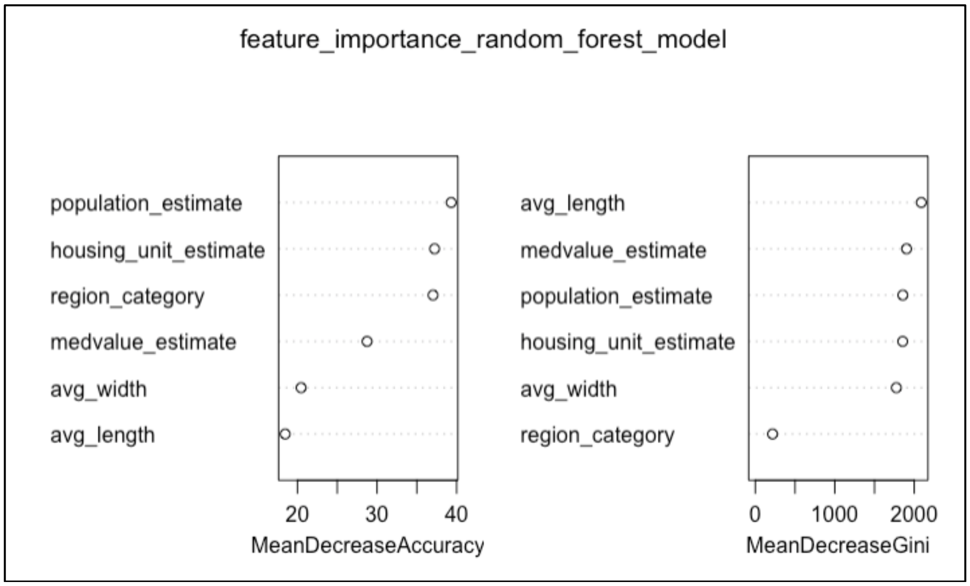

Image 3: Feature Importance of the random forest model utilizing gini importance

Lastly, XGBoost utilized gain importance. The order of rank by gain from highest to lowest includes: average length, median value, population, average width, housing unit, and region category.

Feature	Gain	Cover	Frequency
<chr>	<dbl>	<dbl>	<dbl>
avg_length	0.21904961	0.220995408	0.248068212
medvalue_estimate	0.21208525	0.237082550	0.198774314
population_estimate	0.19616322	0.203533384	0.191846523
avg_width	0.17565480	0.127700700	0.206501465
housing_unit_estimate	0.14251361	0.190497968	0.130295763
region_categoryD	0.03769673	0.013037572	0.009059419
region_categoryT	0.01683678	0.007152416	0.015454303

Image 4: Feature Importance of the XGBoost model utilizing gain importance

As each model determines importance with a different scale, to compare the feature importance across all three they were normalized and combined to plot the following as a visualization:

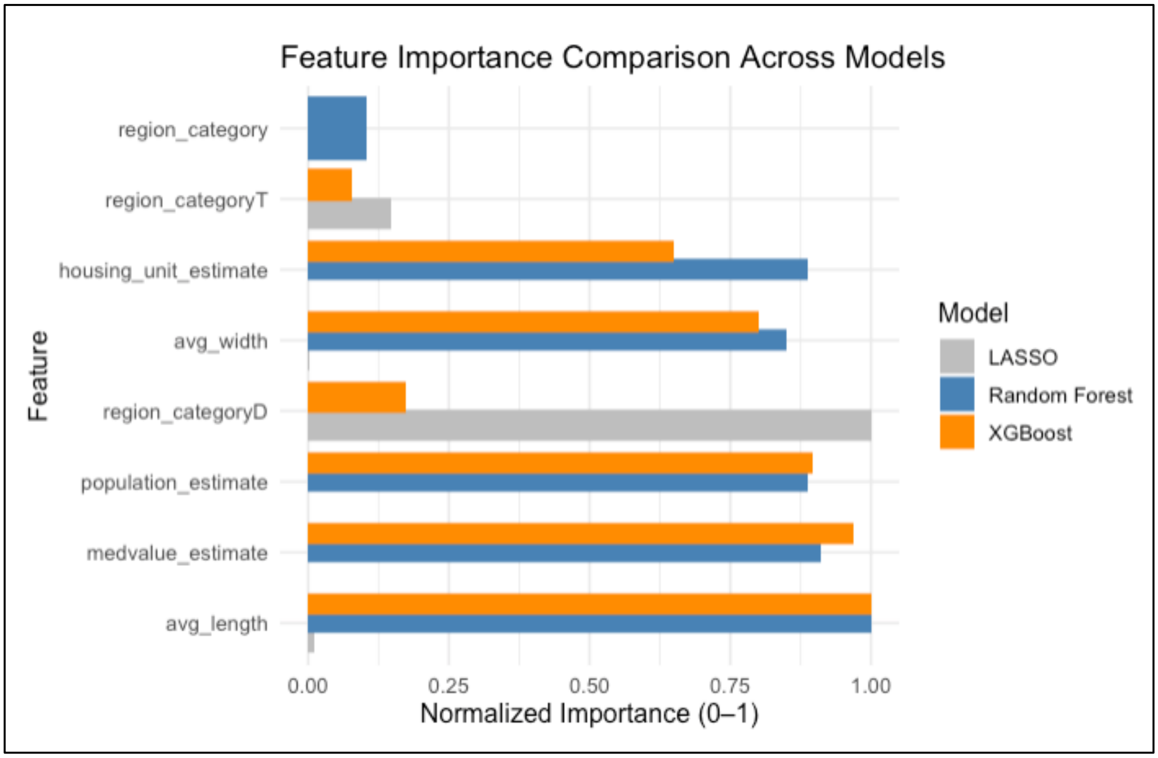

Image 5: Feature Importance comparison across each model (logistic regression, random forest and XGBoost)

Random forest and XGBoost tended to align consistently on feature importance. From all models together, the most important predictor appears to be average length. Median value and population also appeared to be important predictors. Average width and housing unit appear to be moderately important while region is ranked highly in the logistic regression model, however, does not appear to be important in the random forest or XGBoost models. However, I will not be removing any variables due to the importance of retaining region category for the purposes of the research question.

5 Preliminary Model Results

Recall was utilized to examine model sensitivity and PR-AUC was used to determine how well each model performs in ranking risk. The results comparing each are below:

Model	**Recall sensitivity**	**PR-AUC**	**Interpretation**
Logistic Regression	94.20%	0.6861	Over predicts false positive county_year observations as high risk
random forest	52.28%	0.6750	Misses too many high-risk county_year observations
XGBoost	91.15%	0.7075	Most balanced model. Captures most accurate high-risk county_year observations and fewer missed.

Figure 1: Comparing model recall and PR-AUC results

6 Discussion of Results

Logistic regression achieved the highest recall sensitivity of the 3 models, however, it over predicted false positive or high-risk counties. Alternatively, random forest missed many high-risk counties. XGBoost provided the best balance with a high recall sensitivity of 91.15% and the highest PR-AUC at 0.7075. This indicates that it performs the best at accurately predicting high-risk counties, which is most important to the stakeholder.

To improve sensitivity, the target variable hazard cutoff for binary classification was adjusted from the original 75^{th} percentile to the 65^{th} percentile allowing for the top 35% of county-years to be classified as high-risk. This adjustment helped to reduce the risk of missed high-risk observations and aligns with the overall goal to prioritize risk identification.

Feature importance results showed that XGBoost and random forest aligned with most variables and indicated that the most important variables included tornado characteristics.

Next steps include hyperparameter tuning using cross-validation (randomized search) to optimize the model performance and reduce the risk of overfitting. These will include adjusting the number of trees in random forest, depth in xgboost and lambda in Lasso. This will mainly focus on the XGBoost model as this performed the most balanced out of the 3.

7 GitHub

https://github.com/jill-baker/tornado-insurance-risk-repo

8 Appendix

The Data Dictionary below includes identification of key raw, engineered and target variables in the final county-year tornado hazard and insurance exposure dataset. The target variable, high_tornado_hazard_next_year, was created from the hazard-related variables in the dataset: tornado_count_county_year, average_intensity_county, max_intensity_county, total_property_loss_year and strong_tornado_count.

Column Name	Example	Data Type	Description
om	'192'	Integer	Tornado number. Count of tornadoes during given year from NOAA dataset
yr	'1950'	Integer	Year of tornado event (1950-2024)
mo	'3'	Integer	Month of tornado event, (1-12)
dy	'26'	Integer	Day of tornado event, (1-31)
date	'1950-03-26'	Date	Date of tornado event in yyy-mm-dd format
st	'AR'	String	State (two-letter postal abbreviation) of tornado event
mag	'2'	Integer	F-Scale (EF-Scale after 01/07) Values 0, 1, 2, 3, 4, 5, -9 (-9=unknown)
loss	'4'	Numeric	Estimated property loss in millions of dollars. 0 or blank = unknown, not $0
len	'17.4'	Numeric	Length of tornado in miles
wid	'150'	Integer	Width of tornado in yards
county_fips	'05019'	String	Federal Information Processing Standards code for county, full 5 digit code (first 2 digits for state, last 3 digits for county).

county_name_state	'Clark County, Arkansas'	String	Full county name and state
population_estimate	'21125'	Integer	**Predictor.** Present-day exposure variable. Estimated total population in county
housing_unit_estimate	'10029'	Integer	**Predictor.** Present-day exposure variable. Estimated total number of housing units in the county
medvalue_estimate	'155000'	Numeric	**Predictor.** Present-day exposure variable. Estimated median home value in the county (in dollars)
region_category	'T'	String	**Predictor.** Indicates if county is located in Dixie Alley, Tornado Alley or Other. Dixie Alley = D, Tornado Alley = T, Other = O.
high_tornado_hazard_next_year	'1'	Binary	**Target variable.** Indicates whether a county is classified as high tornado hazard in the following year based on county's forward-shifted composite hazard classification (1=high tornado hazard, 0=not high tornado hazard)
tornado_count_county_year	'4'	Integer	*Target creation hazard variable.* Total number of tornadoes per county for a given year
average_intensity_county	'3'	Numeric	*Target creation hazard variable.* Average tornado intensity (F/EF-Scale) for tornadoes in county for a given year.
max_intensity_county	'5'	Integer	*Target creation hazard variable.* The maximum/highest observed tornado intensity per county in given year
total_property_loss_year	'5'	Numeric	*Target creation hazard variable.* The total property loss with all occurred tornadoes per county in given year
average_property_loss	'4'	Numeric	Hazard variable. The average property loss for tornadoes in county for a given year
strong_tornado_count	'3'	Integer	*Target creation hazard variable.* Count of tornadoes in county-year where the tornado intensity (mag) is >=3
property_loss_per_tornado	'50000'	Numeric	Calculation of total_property_loss_year / tornado_count_county_year
tornadoes_per_100k_population	'100'	Numeric	Normalized measure: Calculation of tornado_count_county_year / population_estimate * 100000
tornadoes_per_10k_housing_units	'50'	Numeric	Normalized measure: Calculation of tornado_count_county_year / housing_unit_estimate * 10000

avg_length	'5'	Numeric	**Predictor.** Average length in miles for tornadoes per county
avg_width	'5'	Numeric	**Predictor.** Average width in yards for tornadoes per county
composite_hazard_score	' -0.50'	Numeric	Composite index created by averaging standardized tornado frequency, intensity, and loss variables. Used to classify high hazard counties.
high_tornado_hazard	'1'	Binary	Variable used to create the predictive target high_tornado_hazard_next_year and represents the current-year classification
log_property_loss	'1'	Numeric	Logged transformation of total_property_loss_year
log_population	'1'	Numeric	Logged transformation of population_estimate
log_housing	'1'	Numeric	Logged transformation of housing_unit_estimate
z_avg_intensity	'1'	Numeric	Z-Score from average_intensity_county for target variable creation: high_tornado_hazard_next_year
z_max_intensity	'1'	Numeric	Z-Score from max_intensity_county for target variable creation: high_tornado_hazard_next_year
z_strong_count	'1'	Numeric	Z-Score from strong_tornado_count for target variable creation: high_tornado_hazard_next_year
z_property_loss	'1'	Numeric	Z-Score from log_property_loss for target variable creation: high_tornado_hazard_next_year
z_tornado_count	'1'	Numeric	Z-Score from tornado_count_county_year for target variable creation: high_tornado_hazard_next_year
strong_tornado_fraction	'1'	Numeric	Calculation of strong tornado count divided by total tornado count per county year

9 References & Data Sets

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